Integrated Onion and Tomato Mandi Price Intelligence Report

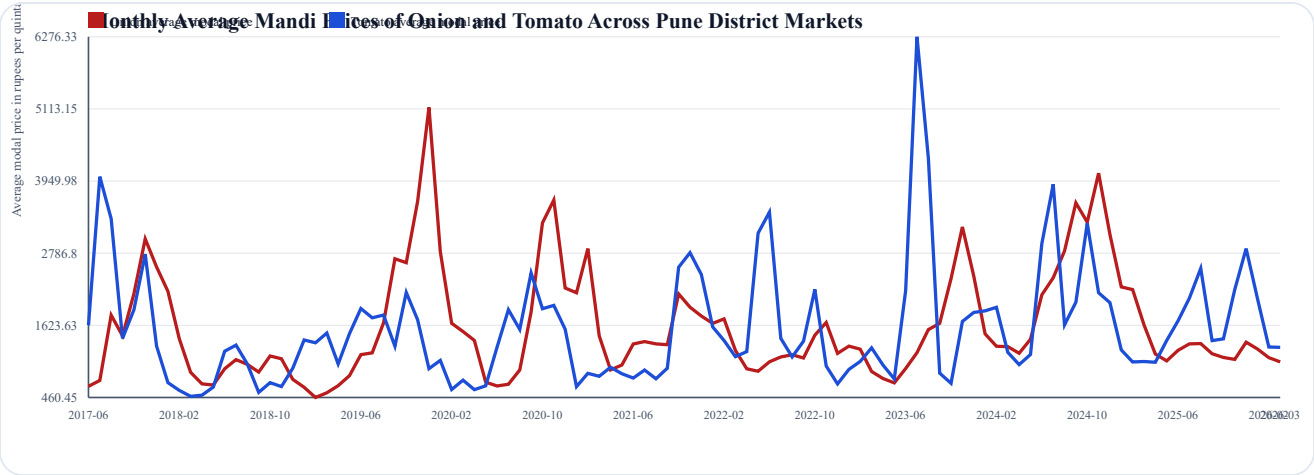
This report combines Pune district onion and tomato mandi prices with nearby district weather information and Mumbai petrol and diesel prices. It covers descriptive analysis, exploratory analysis, correlation review, diagnostic analysis, predictive modeling, future forecasting, and prescriptive action guidance.

Executive Summary

Integrated common dates 3085	**Onion model R2** 0.9371	**Onion model RMSE** 247.88
Tomato model R2 0.894	**Tomato model RMSE** 287.84	**Onion CV lambda** 0.1
Tomato CV lambda 0.1		

1. Onion uses longer lag windows because it is less perishable and retains temporal memory for longer periods.
2. Tomato uses shorter lag windows because the price reacts more quickly to fresh arrivals and spoilage pressure.
3. Ridge regression plus time-series cross-validation was used to control collinearity risk and test model robustness.

Exploratory Data Analysis


Monthly Average Mandi Prices of Onion and Tomato Across Pune District Markets
Onion
Tomato
Average modal price in rupees per quintal
6276.33
5113.15
3949.98
2786.8
1623.63
460.45
2017-06
2018-02
2018-10
2019-06
2020-02
2020-10
2021-06
2022-02
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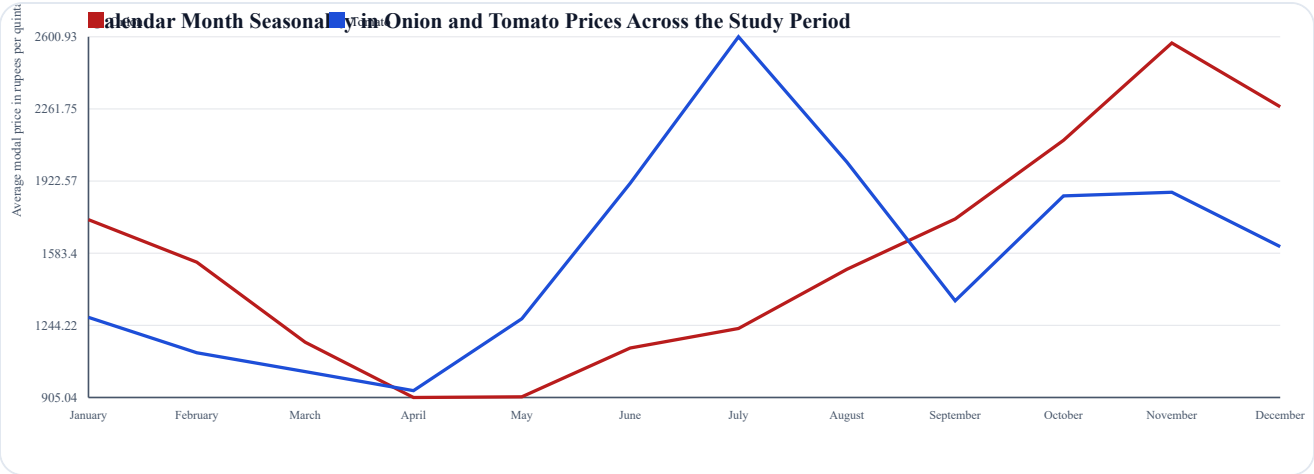
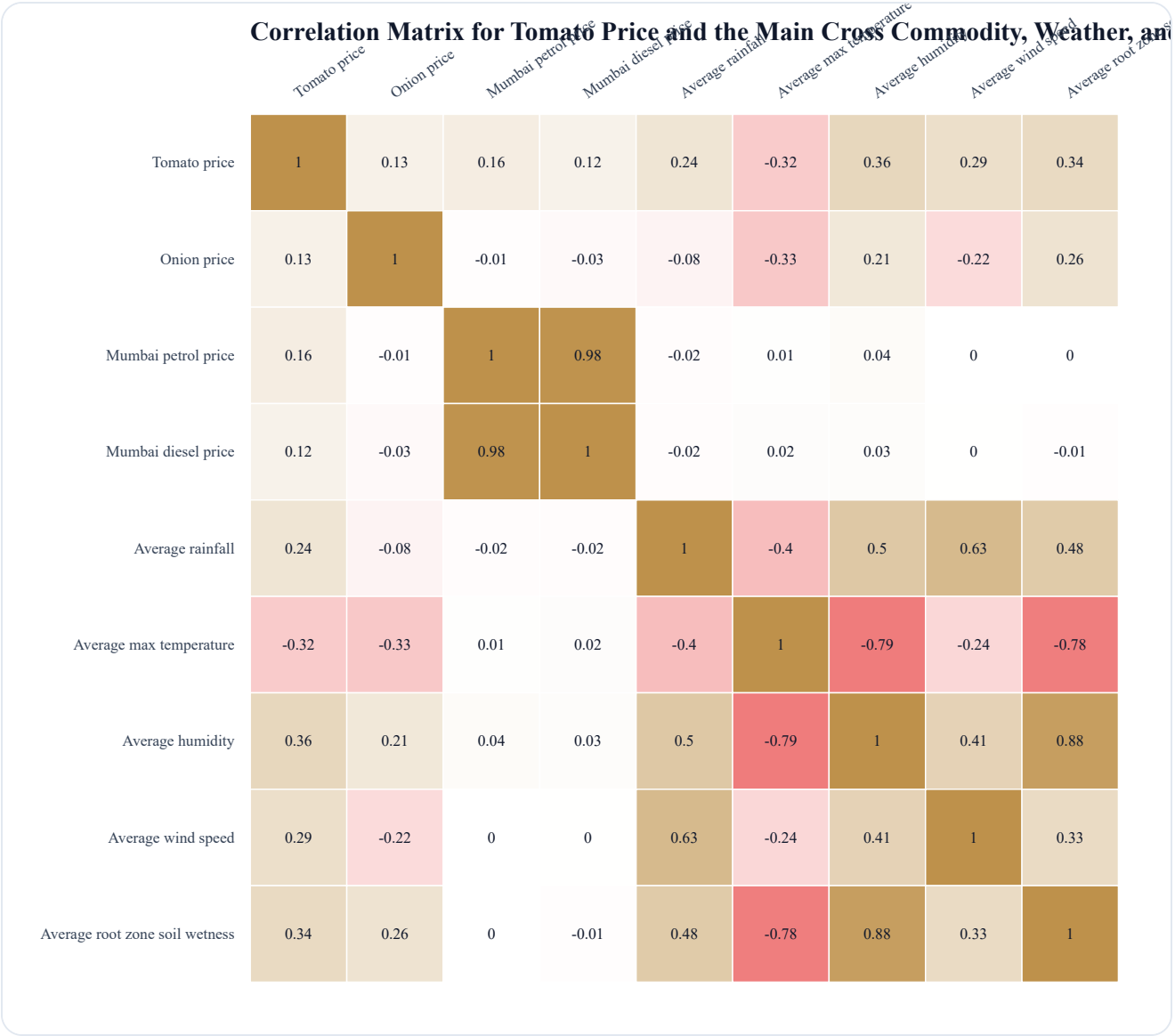
2026-02
2026-03

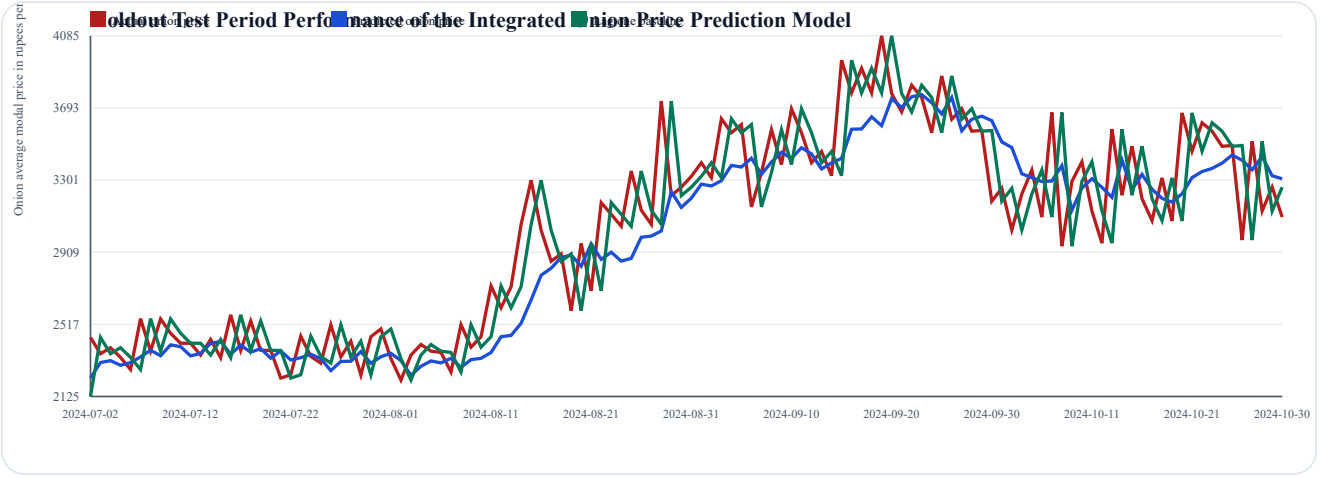

Calendar Month Seasonality in Onion and Tomato Prices Across the Study Period
Onion
Tomato
Average modal price in rupees per quintal
2600.93
2261.75
1922.57
1583.4
1244.22
905.04
January
February
March
April
May
June
July
August
September
October
November
December

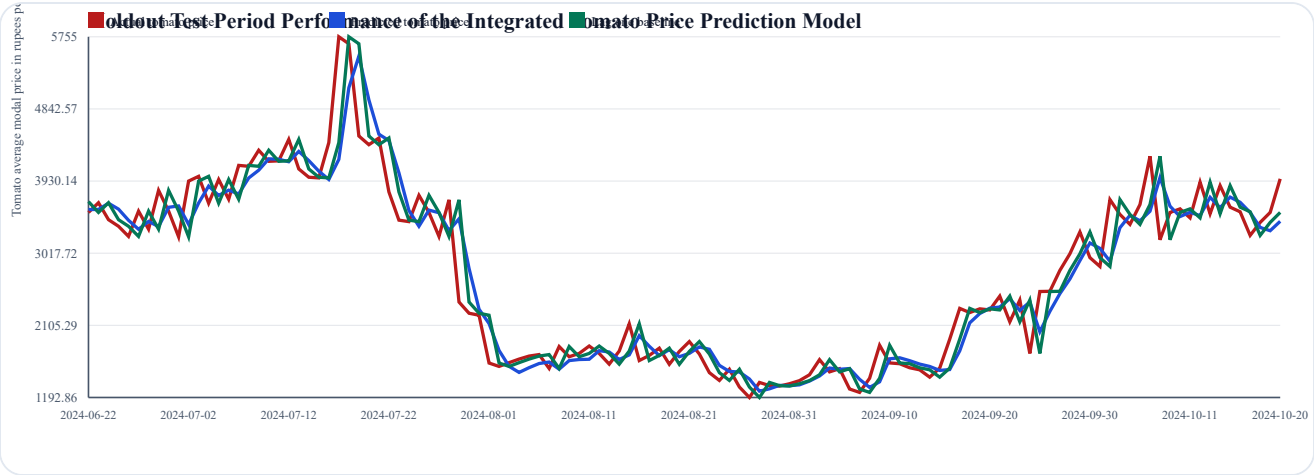
Correlation Matrix for Onion Price and the Main Cross Commodity, Weather, and

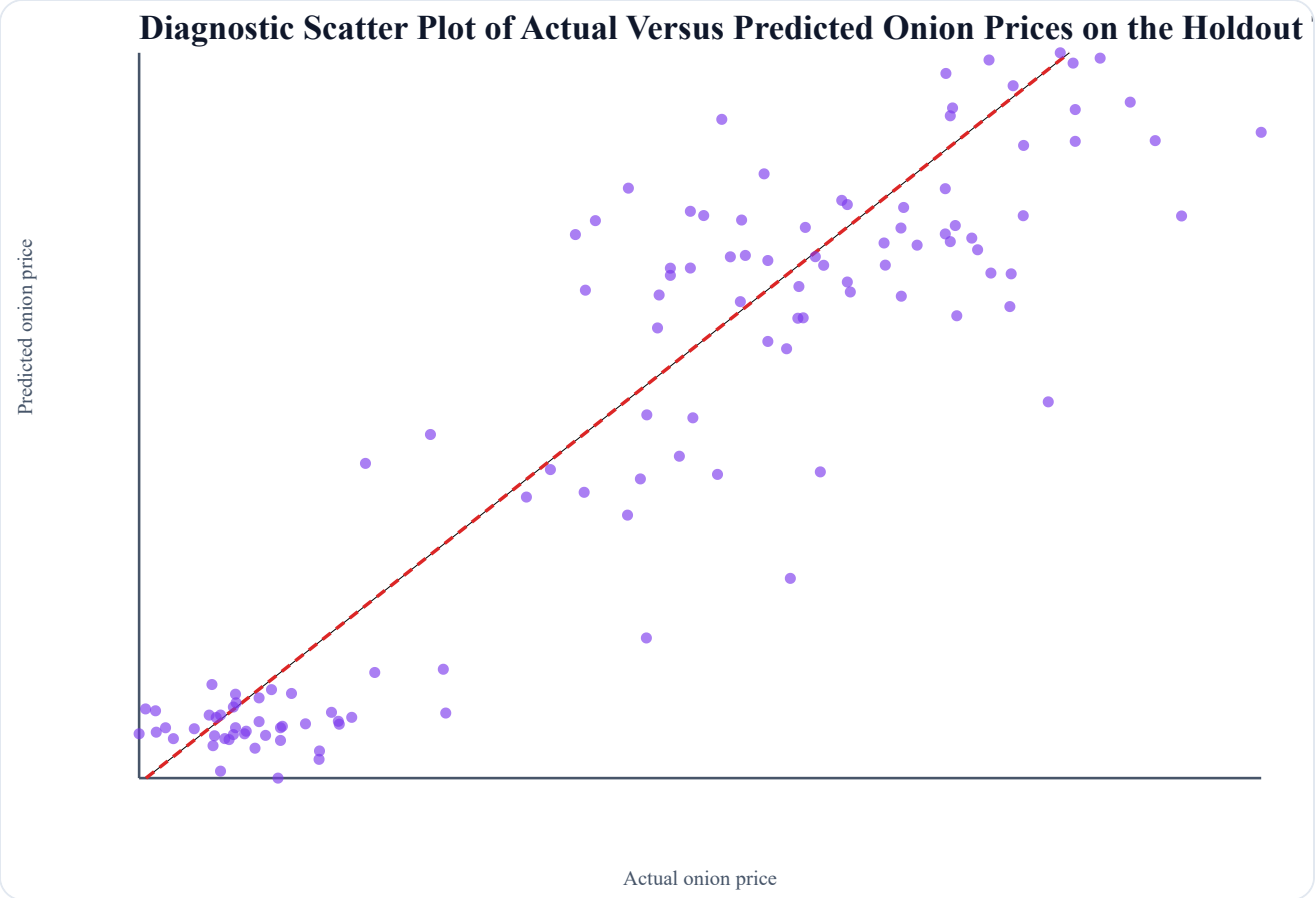
	Onion price	Tomato price	Mumbai petrol price	Mumbai diesel price	Average rainfall	Average max temperature	Average humidity	Average wind speed	Average root zone soil wetness
Onion price	1	0.13	-0.01	-0.03	-0.08	-0.33	0.21	-0.22	0.26
Tomato price	0.13	1	0.16	0.12	0.24	-0.32	0.36	0.29	0.34
Mumbai petrol price	-0.01	0.16	1	0.98	-0.02	0.01	0.04	0	0
Mumbai diesel price	-0.03	0.12	0.98	1	-0.02	0.02	0.03	0	-0.01
Average rainfall	-0.08	0.24	-0.02	-0.02	1	-0.4	0.5	0.63	0.48
Average max temperature	-0.33	-0.32	0.01	0.02	-0.4	1	-0.79	-0.24	-0.78
Average humidity	0.21	0.36	0.04	0.03	0.5	-0.79	1	0.41	0.88
Average wind speed	-0.22	0.29	0	0	0.63	-0.24	0.41	1	0.33
Average root zone soil wetness	0.26	0.34	0	-0.01	0.48	-0.78	0.88	0.33	1

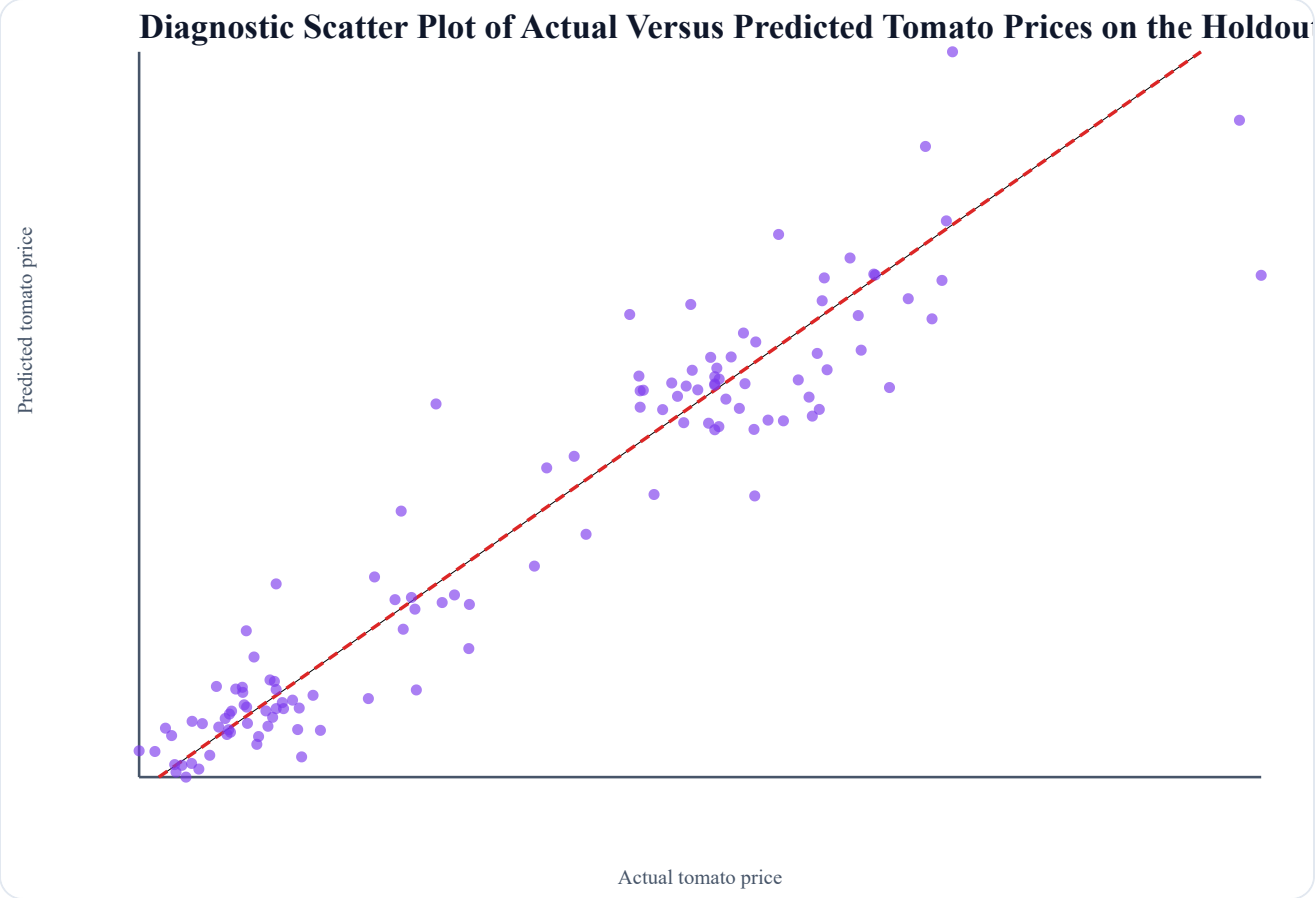


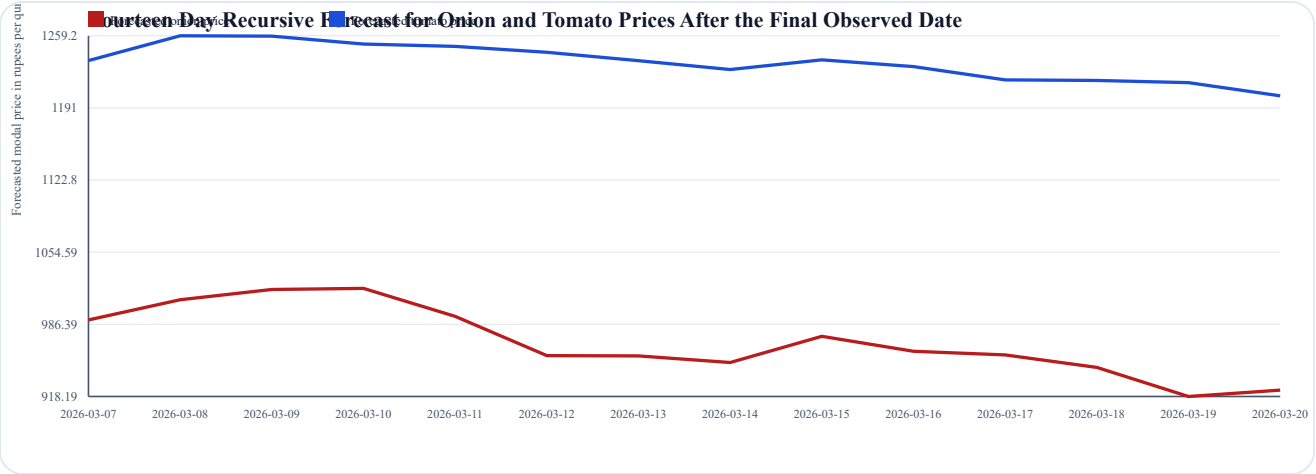
Model Diagnostics and Forecasting




Holdout Test Period Performance of the Integrated Tomato Price Prediction Model
Tomato average modal price in rupees per kg
5755
4842.57
3930.14
3017.72
2105.29
1192.86
2024-06-22
2024-07-02
2024-07-12
2024-07-22
2024-08-01
2024-08-11
2024-08-21
2024-08-31
2024-09-10
2024-09-20
2024-09-30
2024-10-11
2024-10-20


Diagnostic Scatter Plot of Actual Versus Predicted Onion Prices on the Holdout
Predicted onion price
Actual onion price





Prescriptive Analysis

- Onion is forecasted to soften over the next two weeks, so traders should consider faster release, especially if holding costs and shrinkage risk increase.
- Tomato is forecasted to weaken or remain fragile, so the prescriptive action is to prioritize immediate harvesting, sorting, and dispatch to reduce spoilage losses.

- Current low rainfall conditions suggest transport friction is not the main constraint, so price planning can focus more on demand and arrivals.
- Higher Mumbai fuel prices imply transport cost pressure, so grouping shipments and optimizing route density can help protect margins.